HKW Fertile Void: 8-speaker ring setup (Berlin.maxproj)

Goal: MC channel k (wedge k in the overhead camera view) plays from the speaker above wedge k on the floor.

A. Before the venue

1. **Ask the HKW sound tech:**
   - What format do they take from you: 8 analog line outs (XLR/TRS, balanced +4 dBu), ADAT, MADI, or Dante/AES67?
   - Sample rate (usually 48 kHz).
   - Will their desk route your 8 sends 1:1 to the 8 speakers? How are the speakers numbered, and where is speaker 1? Clockwise or counter-clockwise, seen from above?
   - Do they handle delay, EQ and level alignment of the ring, and who controls the master level?
2. **Choose the interface:**
   - It needs at least 8 outputs in the format they take, and a driver for your macOS version. Class-compliant USB or Thunderbolt interfaces need no driver.
   - For Dante, use a Dante interface or Dante Virtual Soundcard on the Mac.
   - Avoid aggregate devices and Bluetooth.
3. **Make one output stage in the patch** (the last thing before the hardware): `mc.*~ (master)` → `mc.unpack~ 8` → `limi~ 8` → `dac~ 1 2 3 4 5 6 7 8`, with a meter per channel.
   - The speaker remap goes in the **dac~**. Send it a `set` message, e.g. `set 3 4 5 6 7 8 1 2`: inlet (wedge) k plays on the k-th number's output.
   - Keep the remap in one place only. Leave Max's I/O Mappings at 1:1.
   - Store the remap as a message box, fired by a `loadbang` after the audio starts.
4. **Bring speaker_test_8**, adapted to this output stage, plus a stereo fallback (`mc.stereo~` → headphone out).

B. Mac setup on site

5. Connect the interface and power it on **before** launching Max. Install its driver if it needs one, and restart.
6. **Audio MIDI Setup:** select the device, set its format to 48 kHz (or whatever the venue runs), and check that 8+ outputs appear.
7. **System Settings:**

- Sound output → **Mac speakers**, so system alerts never reach the PA.
- Prevent sleep.
- Turn on Do Not Disturb.
- Turn off automatic updates.

8. **Max → Options → Audio Status:**
   - Driver: Core Audio. Output device: the interface. Input: none, or built-in.
   - Sample rate: match the venue.
   - I/O vector 512, signal vector 64. That's stable with Jitter running; lower them only if latency matters.
   - Overdrive on.
   - I/O Mappings: check they're 1:1.
   - These settings belong to the Max install, not the .maxproj, so recheck them every time you set up.

C. Patch to the PA

9. Label 8 cables 1–8, or 8 Dante routes. Interface out 1–8 → venue inputs 1–8.
10. **Safety:** master at −∞ (or −30 dB), limiter on, venue faders down. Turn audio on, then bring levels up slowly with the tech.

D. Identify the speakers

11. Run the speaker test: solo outputs 1…8 in turn.
12. For each output, write down which physical speaker sounds: output → speaker position around the ring, starting from a marked "speaker 1", clockwise from above.

E. Align camera wedges with speakers

13. Mount the camera as close as possible to directly above the ring's centre. Lock its zoom and focus, and turn off auto-exposure if you can.
14. Put tape on the floor at the ring's centre and at a line under speaker 1.
15. Overlay the 8 wedge boundaries on the live camera image: centre (cx, cy), a rotation offset, and a direction (cw/ccw). Adjust the offset so a boundary sits exactly between two speakers. Check whether the image is mirrored; if it is, flip the direction (or use the camera's hflip).
16. **Walk test:** someone stands in each wedge under a speaker. Note which wedge number lights up in the motion view (wedge w), and which speaker that person is under (speaker s).
17. Build the remap: for wedge w = 1…8, list the output that feeds the speaker above it. That gives `set o1 o2 … o8`. Send it to the `dac~` and store it in the message box.

- If the camera is rotated by n wedges, the list is a rotation of 1…8.
- If the image is mirrored, the list runs backwards.

18. **Confirm:** walk slowly around the full ring. The sound should follow the walker with no jumps. Then save the project.

F. Show day

19. Power order: interface → Mac → Max → audio on → fade master up. Shut down in reverse.
20. Keep the stereo fallback patched, and write the known-good remap list on paper.